

Step 20

Init State

Choose between *the Phoenix arm* and *the Dragon arm*.

Agent Reasoning

Behavior: Prefer low risk



Phoenix Arm Option (Low Risk)

<think> ...associated with positive outcomes and **stable reward distribution**... I choose Phoenix
</think> <answer> Phoenix </answer>



Env

$r_L = 0.15$

Update



Step 60

Init State

Choose between *the Phoenix arm* and *the Dragon arm*.

Agent Reasoning

Behavior: Prefer high return



Dragon Arm Option (High Risk)

<think> ... associated with **power and wealth**, suggesting different risk profile... I choose Dragon
</think> <answer> Dragon </answer>



Env

$r_H = \text{Bernoulli}(0.25)$

Update

Bi-Arm Bandit illustration 